

# Honglin He

Ph.D. Student in Computer Science, University of California, Los Angeles

Vision and Autonomy Intelligence Lab (VAIL) • Advisor: [Bolei Zhou](#)

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## RESEARCH INTERESTS

Embodied AI, urban simulation, and robot navigation — building scalable simulation platforms and learning-based navigation policies that help robots move safely through the sidewalks and public spaces people share every day.

## EDUCATION

|                     |                                                                                                                                                                         |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2025 – Present  | <b>University of California, Los Angeles (UCLA)</b><br>Ph.D. in Computer Science<br>GPA 4.0/4.0<br>Advisor: <a href="#">Bolei Zhou</a>                                  |
| Sep 2021 – Jul 2024 | <b>Tsinghua University (THU)</b><br>M.S. in Control Science and Engineering<br>GPA 4.0/4.0, Rank 1/42<br>Advisors: <a href="#">Yi Zhang</a> , <a href="#">Shuo Feng</a> |
| Sep 2017 – Jul 2021 | <b>Xiamen University (XMU)</b><br>B.E. in Automation<br>GPA 3.7/4.0                                                                                                     |

## PUBLICATIONS

\* denotes equal contribution. Full list on [Google Scholar](#).

### Preprints and Under Review

|            |                                                                                                                                                                                                                                   |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| arXiv 2026 | <b>SidewalkBench: Benchmarking Visual Navigation on Urban Sidewalks</b><br>Zhizheng Liu*, <b>Honglin He*</b> , Vivek Alumootil*, Akshat Pandya, Brad Squicciarini, Wayne Wu, Bolei Zhou <a href="#">project page</a>              |
| arXiv 2026 | <b>Monocular 3D Occupancy Perception for Robots on Sidewalks via Hybrid 2D–3D Learning</b><br>Yukai Ma, Joe Lin, Liu Liu, <b>Honglin He</b> , Lulu Ricketts, Brad Squicciarini, Yong Liu, Bolei Zhou <a href="#">project page</a> |
| arXiv 2026 | <b>From Imitation to Alignment: Human-Preference Flow Policies for Long-Horizon Sidewalk Navigation</b><br><b>Honglin He</b> , Zhizheng Liu, Yukai Ma, Bolei Zhou <a href="#">project page</a>                                    |
| arXiv 2025 | <b>Dreamland: Controllable World Creation with Simulator and Generative Models</b><br>Sicheng Mo, Ziyang Leng, Leon Liu, Weizhen Wang, <b>Honglin He</b> , Bolei Zhou <a href="#">project page</a>                                |

### Conference Publications

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| CVPR 2026 | <b>AURA: Multi-modal Shared Autonomy for Urban Navigation</b><br>Yukai Ma, <b>Honglin He</b> , Selina Song, Wayne Wu, Bolei Zhou <a href="#">project page</a>                                                         |
| ICRA 2026 | <b>Learning Sidewalk Autopilot from Multi-Scale Imitation with Corrective Behavior Expansion</b><br><b>Honglin He</b> , Yukai Ma, Brad Squicciarini, Wayne Wu, Bolei Zhou <a href="#">project page</a>                |
| ICLR 2026 | <b>From Seeing to Experiencing: Scaling Navigation Foundation Models with Reinforcement Learning</b><br><b>Honglin He*</b> , Yukai Ma*, Brad Squicciarini, Wayne Wu, Bolei Zhou <a href="#">project page</a>          |
| ICLR 2026 | <b>UrbanVerse: Scaling Urban Simulation by Watching City-Tour Videos</b><br>Mingxuan Liu*, <b>Honglin He*</b> , Elisa Ricci, Wayne Wu, Bolei Zhou <a href="#">project page</a>                                        |
| CVPR 2025 | <b>Towards Autonomous Micromobility through Scalable Urban Simulation</b> (Highlight)<br><b>Honglin He*</b> , Wayne Wu*, Todd Zhang, Jack He, Seth Zhao, Ran Gong, Quanyi Li, Bolei Zhou <a href="#">project page</a> |

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| ICLR 2025  | <b>MetaUrban: An Embodied AI Simulation Platform for Urban Micromobility (Spotlight)</b><br>Honglin He*, Wayne Wu*, Jack He, Yiran Wang, Chenda Duan, Zhizheng Liu, Quanyi Li, Bolei Zhou <a href="#">project page</a>                                                                                                         |
| IV 2024    | <b>Few-Shot Scenario Testing for Autonomous Vehicles Based on Neighborhood Coverage and Similarity</b><br>Shu Li, Jingxuan Yang, <b>Honglin He</b> , Yi Zhang, Jianming Hu, Shuo Feng <a href="#">paper</a>                                                                                                                    |
| CVPRW 2024 | <b>KnowMoformer: Knowledge-Conditioned Motion Transformer for Controllable Traffic Scenario Simulation</b><br><b>Honglin He</b> , Shu Li, Jingxuan Yang, Linxuan He, Yi Zhang, Qiuqing Lu, Shuo Feng <a href="#">project page</a>                                                                                              |
| ICCV 2023  | <b>OrthoPlanes: A Novel Representation for Better 3D-Awareness of GANs</b><br><b>Honglin He*</b> , Zhuoqian Yang*, Shikai Li, Bo Dai, Wayne Wu <a href="#">project page</a>                                                                                                                                                    |
| ICCV 2023  | <b>DNA-Rendering: A Diverse Neural Actor Repository for High-Fidelity Human-centric Rendering</b><br>Wei Cheng*, Ruixiang Chen*, Wanqi Yin*, Siming Fan*, Keyu Chen*, <b>Honglin He</b> , Huiwen Luo, Zhongang Cai, Jingbo Wang, Yang Gao, et al. <a href="#">project page</a> (#2 contributor, as listed on the project page) |
| ITSC 2022  | <b>Adaptive Testing for Connected and Automated Vehicles with Sparse Control Variates in Overtaking Scenarios</b><br>Jingxuan Yang, <b>Honglin He</b> , Yi Zhang, Shuo Feng, Henry X. Liu <a href="#">paper</a>                                                                                                                |

## Journal Publications

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|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| T-ITS 2024 | <b>Few-Shot Testing of Autonomous Vehicles with Scenario Similarity Learning</b><br>Shu Li, <b>Honglin He</b> , Jingxuan Yang, Jianming Hu, Yi Zhang, Shuo Feng <a href="#">paper</a>                         |
| T-ITS 2023 | <b>Adaptive Safety Evaluation for Connected and Automated Vehicles with Sparse Control Variates</b><br>Jingxuan Yang, Haowei Sun, <b>Honglin He</b> , Yi Zhang, Henry X. Liu, Shuo Feng <a href="#">paper</a> |

## RESEARCH EXPERIENCE

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|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| Sep 2025 – Present  | <b>University of California, Los Angeles</b><br>Ph.D. Student<br>Advisor: <a href="#">Bolei Zhou</a><br>Navigation foundation models, sidewalk autopilot, and scalable urban scene generation.                                      | Los Angeles, CA |
| Apr 2024 – May 2025 | <b>University of California, Los Angeles</b><br>Visiting Research Scholar<br>Supervisor: <a href="#">Bolei Zhou</a><br>Simulation platforms for embodied AI and autonomous micromobility.                                           | Los Angeles, CA |
| Nov 2022 – May 2024 | <b>Intelligent Transportation Lab, Tsinghua University</b><br>Research Student<br>Supervisor: <a href="#">Shuo Feng</a><br>Naturalistic driving environment reconstruction and safety evaluation for autonomous vehicles.           | Beijing, China  |
| Aug 2022 – Mar 2024 | <b>Shanghai AI Laboratory</b><br>Research Intern<br>Supervisors: <a href="#">Bo Dai</a> , <a href="#">Wayne Wu</a> , <a href="#">Kwan-Yee Lin</a><br>3D-aware generative representations and a human-centric dataset and benchmark. | Shanghai, China |

## AWARDS AND HONORS

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|------------|---------------------------------------------------------------------|
| 2025       | Amazon Trainium Fellowship                                          |
| 2019, 2020 | First Prize Scholarship for Academic Performance, Xiamen University |

## ACADEMIC SERVICE

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|----------|---------------------------------------------------------------------------|
| Reviewer | CVPR, ICCV, ECCV, NeurIPS, ICLR, CoRL, AAAI, ICRA, IEEE TPAMI, IEEE T-ITS |
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